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28 September 1984
SUPERSEDING
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16 June 1982

MILITARY SPECIFICATION

HEATER, WATER, LIQUID FUEL; M-80

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a portable field water heater capable of supplying heated water, 100°F rise at a rate of 9 gallons per minute. Alternating current 208 volts electric power is required and any type gasoline or diesel fuel may be used.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-E-529	-	Enamel, Alkyd, Semi-Gloss
TT-I-1795	-	Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces)
TT-W-572	-	Wood-Preservative: Water-Repellent
VV-G-1690	-	Gasoline, Automotive, Leaded or Unleaded

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4520

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PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-T-60 - Tape: Packaging, Waterproof

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-B-121 - Barrier Material, Greasproofed, Waterproofed, Flexible
MIL-O-2104 - Lubricating Oil, Internal Combustion Engine, Tactical Service
* MIL-T-22085 - Tape, Pressure-Sensitive, Adhesive, Preservation and Sealing

STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking for U.S. Military Property

* DRAWINGS

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

6-1-6200 - Heater Assembly, Water, M-80
6-1-6203 - Section Assembly, Vessel Outside
6-1-6210 - Firebox Assembly

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting officer or as directed by the contracting officer.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

UNDERWRITERS LABORATORIES, INC.

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY

Standard Qualification Procedure of the American Welding Society

(Application for copies should be addressed to American Welding Society, 1501 Northwest 7th Street, Miami, FL 33125.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Welding Qualifications of the American Society of Mechanical Engineers

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70 - National Electric Code

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02169.)

HYDRAULIC INSTITUTE STANDARDS

(Application for copies should be addressed to the Standard Hydraulic Institute, 122 East 42nd Street, New York, NY 10017.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

* 3.1 Alternate components. Components offered as equivalent to components specified herein and on the referenced drawings as a specific manufacturer's part number or equal shall be functionally equal with the manufacturer's part number identified. The incorporation of such a component in the design of the specified end product shall not require modification to any components, unless such modification is specifically approved by the contracting officer. Prior to

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commencing production, the contractor shall submit for the contracting officer's approval, a list identifying each alternate component, together with proof that each listed component complies with requirements specified herein. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data does not relieve the contractor of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.2 First article. When specified a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.3 Materials and components. The materials and components shall be as specified herein and on the applicable drawings. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all the provisions of this document (see 6.4).

3.3.1 UL standard and certificate. The flame safeguard control shall be UL approved. Prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative that the unit meets the UL requirements. Acceptable evidence shall be a UL listing mark or certification from an independent testing laboratory acceptable to the contracting officer or his authorized representative.

* 3.4 Design and construction. The design and construction of the heater shall be as specified herein, and as shown on Drawing 6-1-6200 and all subsidiary drawings and parts lists pertaining thereto. The heater shall operate on 3 phase, 60 hertz, 208 volts, 5 wire electric power source. Three wires are to supply the phases, the white wire is to be 115 volt neutral ground and the green wire is to be the equipment ground. All wiring shall be in accordance with the provisions of NFPA No. 70. Diesel shall be the primary fuel for operation of the heater and shall be in accordance with VV-F-800. Gasoline shall be the secondary fuel and shall be in accordance with VV-G-1690 and when used lubricating oil (MIL-O-2104, OE-30) shall be added at a 1 to 20 ratio (one quart for 5 gallons of gasoline).

3.5 Performance.

3.5.1 Heater.

3.5.1.1 Firebox assembly. Each firebox assembly (Drawing 6-1-6210) shall have no visible evidence of leakage when tested as specified in 4.5.1.

3.5.1.2 Outside vessel section. Each outside vessel section (Drawing 6-1-6203) shall have no visible evidence of leakage when tested as specified in 4.5.2.

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3.5.1.3 Water heater assembly. Each water heater assembly (Drawing 6-1-6200) shall have no visible evidence of leakage when tested as specified in 4.5.3.

3.5.2 Controls.

3.5.2.1 Water level control. The water level control shall prevent operation of the burner and sound the buzzer whenever the water level in the heater is below the end of the water level electrode, when tested as specified in 4.5.4. Also the burner shall restart and the buzzer stop when the water level is raised to the operating condition, when tested as specified in 4.5.4.

* 3.5.2.2 Operating control. The operating control shall be located in the probe box and wired in series with the limit control. The control shall be provided with a calibrated knob and dial in degrees of Farenheit. The control shall be adjustable over a temperature range of 0° to 250°F. The control shall operate the burner so that the outlet water temperature is maintained between 175° and 210°F (see 4.5.5).

3.5.2.3 Limit control. The limit control shall be located in the upper manifold and wired in series with the operating control. The control shall be factory preset at 190°F and shall be normally closed and shall open on temperature rise. The control shall operate the burner so that the outlet water temperature is maintained between 180° and 194°F (see 4.5.6).

3.5.3 Flame safeguard. The flame safeguard shall operate between 125°F and -20°F. The scanner component shall operate between 212°F and -40°F. Prior to each start the control shall provide a 7-second purge cycle. Following each purge cycle, a 10-second trial for ignition shall occur. When flame failure occurs the control shall de-energize the solenoid, shut off the burner motor and ignition transformer, provide a mechanical lockout and turn on the alarm circuit.

* 3.5.3.1 Flame failure, initial start lockout. Flame failure initial start lockout shall be:

a. a 7-second purge.
b. A 10-second trial for ignition when the solenoid valve is open and then closed.
c. A lockout shall occur in not greater than 60 seconds.

When tested as specified in 4.5.7, lockout time shall be not greater than 90 seconds.

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3.5.3.2 Firing period lockout. The firing period lockout requirements shall be:

a. A 4-second response cycle, when the solenoid valve closes.
b. A 10-second purge cycle.
c. A 10-second trial for ignition when the solenoid is opened and then closed.
* d. Not greater than a 60-second mechanical lockout occurrence.

When tested as specified in 4.5.8, lockout time shall be not greater than 90 seconds.

3.5.3.3 Signal flame. The flame control shall not be effected by the ignition spark and shall have a failure response of less than 2 seconds (see 4.5.9).

3.5.4 Signal device. The signal device shall sound when electric power is applied and the water level control or flame safeguard is deactivated (see 4.5.10).

3.5.5 Circuit breaker. All the circuit breakers shall operate without damage to themselves or other electrical components (see 4.5.11).

* 3.5.5.1 Main breaker. The main circuit breaker shall open within a 3 minute period at a current flow of 8 amps and shall be reset by closing the switch (see 4.5.12).

* 3.5.5.2 Line starter. The line starter shall open within a 3 minute period at a current flow of 8 amps and shall be reset by means of a reset button (see 4.5.13).

3.5.5.3 Burner motor. The burner motor shall be protected by a thermal overload protector that is an integral part of the motor. The motor shall shut off in 25 seconds and be capable of resetting without the use of tools (see 4.5.14).

3.5.6 Valve. The relief valve shall open at a pressure of 60 psi gage and close when the pressure is below 58 psi gage (see 4.5.15).

3.5.7 Burner smoke. There shall be no visible smoke at the pipe stack outlet one minute after ignition of the burner (see 4.5.16).

3.5.8 Heater capacity. The heater shall provide a flow of 9 gallons per minute with a temperature rise of not less than 100°F (see 4.5.17).

3.5.9 Pump, fuel. The fuel pump shall be adjusted to operate at a pressure of 100 psi ± 2 gage (see 4.5.18).

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3.5.10 Heater assembly. The heater assembly shall be put in operation without the use of tools, except for a 7/8 inch wrench to attach the fuel lines to the drum fill assembly (see 4.5.19).

3.5.11 Leakage. The heater assembly shall show no evidence of leakage when tested as specified in 4.5.20.

3.6 Marking. Marking for identification shall be in accordance with MIL-STD-130.

3.7 Interchangeability. All component parts of any one heater shall be so fabricated as to be interchangeable with the like parts of any other unit furnished under any one contract.

3.8 Workmanship. The heater components shall be free from fractures, splits, punctures, tears, dents, creases, bows, miscasts, deteriorations or malformations. There shall be no sharp edges, burrs or slivers. The finish shall be free of blisters, peel, chipped areas or areas of no film. The finish shall be free from smudges, poor touch up, scale, scratches, stain, overspray sag, imprint or embedded foreign matter. All bolts, studs, nuts, and screws shall be drawn tight.

3.8.1 Welders. Unless otherwise specified (see 6.2), before assigning any welder to work, the contractor shall provide the Government inspection service with the names of the welders to be employed in the work, together with certification that each of the welders has passed a qualification test as prescribed by either of the following listed codes for the type of welding operation to be performed, and that such qualification is effective as defined by that particular code as follows:

(a) Standard Qualification Procedure of the American Welding Society.
(b) Welding Qualification of the American Society of Mechanical Engineers.

3.8.2 Welding. The surface of parts to be welded shall be free from oxide, scale, paint, grease, and other foreign matter. Welds shall be sound, continuous, smooth, and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux deposits (when flux is used) shall be removed from the finished welds.

3.8.3 Electrical wiring. Wiring shall not be cut, abraded, or have excessive insulation stripped, and shall be properly and tightly joined at terminals. Wiring shall have adequate slack to provide stain relief.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.2), it shall be examined as specified in 4.4.2.1 and 4.4.2.2 and tested as specified in 4.5.1 thru 4.5.20. The presence of any defects or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 In-process inspection. In-process examination shall be made of surface preparation prior to application of the finished coat to determine compliance with 3.4. Also examination of the surface preparation prior to welding shall be made to determine compliance with 3.8.2. Whenever non-conformance is noted, correction shall be made to the applicable process and all items processed.

4.4.2 End item inspection. The lot shall be expressed in terms of completely assembled water heaters.

4.2.2.1 Visual examination. The end item shall be examined for the defects listed in table I. The sample unit shall be one heater. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

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TABLE I. Visual defects

Examine	Defect	Classification Major	Classification Minor
Finish	Color not as specified		X
	Blistered, peeled, chipped, or area of no film		X
	Poor touch-up, orange peel, color separation, smudge, scale, scratched, stained, overspray, sag, imprint, or foreign matter imbedded		X
Design, construction, and workmanship, general (applicable to all components and assemblies)	Any component or part missing or not specified type or size	X	
	Fractured, split, punctured, torn, dented, creased deteriorated, or malformed	X	
	Component misplaced or not in proper alignment		X
	Component loose		X
	Sharp burr or edge or sliver		X
	Any operation omitted or not properly performed	X	
	Any functioning component that is inoperative or will not operate as intended	X	
	Any functioning component that requires abnormal force to operate or not interchangeable		X
	Any adjustable assembly that cannot be adjusted		X
	Any assembly not properly adjusted		X
	Any component not readily accessible for servicing where required		X
Welding	Missing, incomplete, cracked, fractured, not fused	X	
	Undercut, not smooth and uniform, or scale or flux deposits not removed		X
Soldering (when applicable)	Missing, not adherent, or incomplete		X
	Not clean (flux residue not removed, pinholes in solder)		X

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TABLE I. Visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Bolts, nuts, screws, studs, and other types of threaded fasteners	Broken, stripped, fractured, or not tight		X
Rivets (when required)	Broken, not peened or insufficiently peened, or loose		X
Electrical assembly	Wiring cut, abraded, or not properly installed or joined	X	
	Adequate slack not provided for wiring to relieve strain		X
Gauges, instruments, controls, thermometer, etc.	Not properly mounted or not specified type or capacity		X
	Not free of foreign matter		X
Marking	Missing, incomplete, not legible, not specified type or size, not in proper location		X

4.4.2.2 Dimensional examination. The end item shall be examined for compliance with dimensions specified. Any dimension that it not within the specified tolerance shall be classified as a defect. The sample unit shall be one heater. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

4.4.2.3 End item testing. Each production end item shall be tested for the characteristics listed in table II. Failure of any test shall be cause for rejection of the end item involved. All heater function tests 4.5.4, 4.5.5, 4.5.7, 4.5.8, 4.5.10, 4.5.16, 4.5.17, and 4.5.18 may be performed using test supply and return lines in lieu of the fuel lines that will be furnished with each unit.

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TABLE II. Performance tests

Characteristic	Requirement paragraph	Test method
Firebox leakage	3.5.1.1	4.5.1
Outside vessel leakage	3.5.1.2	4.5.2
Water heater leakage	3.5.1.3	4.5.3
Water level control	3.5.2.1	4.5.4
Operating control	3.5.2.2	4.5.5
Initial start lockout	3.5.3.1	4.5.7
Firing period lockout	3.5.3.2	4.5.8
Signal device	3.5.4	4.5.10
Burner smoke	3.5.7	4.5.16
Heater capacity	3.5.8	4.5.17
Fuel pump pressure	3.5.9	4.5.18
Heater assembly	3.5.11	4.5.20

4.4.3 Packaging inspection. An examination shall be made to determine that preservation, packing and marking comply with section 5. Defects shall be scored as set forth in table III. The sample unit shall be one shipping container fully packaged except that it need not be closed. The lot shall be the number of shipping containers offered for inspection at one time. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

TABLE III. Packaging defects

Examine	Defect
Markings	Omitted, incorrect; illegible; of improper size, location, sequence, or method of application
Preservation	Improperly applied or missing
Materials	Components missing or damaged
Workmanship	Inadequate application of components such as incomplete closure of barrier materials, not taped as specified System not completely drained and dried

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4.4.4 Certification of compliance. Certification of welders, UL Flame safeguard and NFPA wiring shall be furnished and acceptable to the Government representative for the first article and all production items.

4.5 Methods of inspection. Commercial grade No. 1 or No. 2 fuel oil can be used as the fuel for the applicable tests.

4.5.1 Firebox assembly test. Each firebox assembly shall be separately tested for leaks under a hydrostatic pressure of 75 psi gage for a period of 5 minutes. Any nonconformance with the requirements of 3.5.1.1 shall constitute failure of this test.

4.5.2 Vessel outside section test. Each outside vessel section shall be separately tested for leaks under a hydrostatic pressure of 75 psi gage for a period of 5 minutes. Any nonconformance with the requirements of 3.5.1.2 shall constitute failure of this test.

4.5.3 Water heater assembly test. Following the heater capacity test (4.5.17) each heater, water assembly shall be separately tested for leaks under a hydrostatic pressure of 75 psi gage for a period of 5 minutes. To examine the firebox for leakage the front cover and burner assembly shall be removed. Any nonconformance with the requirements of 3.5.1.3 shall constitute failure of this test.

4.5.4 Water level control test. The water level control shall be tested with the heater in operating condition. Any nonconformance with the control requirements of 3.5.2.1 shall constitute failure of this test.

4.5.5 Operating control test. The operating control shall be tested with the heater in operating condition. The operating control shall be set at 140°F and the limit control by-passed. The burner shall be operated at 100 psi gage with a 5 GPH nozzle and adjusted below the smoke point. A constant water flow of 5 ± 0.5 gallons per minute thru the heater shall be maintained. After equilibrium has been reached the unit shall be allowed to operate thru three continuous cycles. The temperature setting of the operating control may be adjusted to meet the water outlet temperature requirements in 3.5.2.2. Any nonconformance with the water outlet temperature requirements as specified in 3.5.2.2 shall constitute failure of this test.

4.5.6 Limit control test. The limit control shall be tested with the heater in operating condition. The operating control shall be by-passed and a constant flow of 5 GPM ± .5 maintained thru the heater. The burner shall be operated at 100 psi gage with a 5 GPH nozzle. The burner shall be adjusted slightly below the smoke point. After equilibrium has been reached the unit shall be operated thru three continuous cycles. Any nonconformance with the water outlet temperature requirements in 3.5.2.3 shall constitute failure of this test.

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* 4.5.7 Initial start lockout test. The oil valve shall be closed and the unit tested for initial flame lockout. Any nonconformance with the lockout time of not more than 90 seconds as specified in 3.5.3.1 shall constitute failure of this test.

* 4.5.8 Firing period lockout test. The burner shall be operated for more than 2 minutes. The oil valve shall be closed and the unit shall be tested for flame failure during the firing period. Any nonconformance with the lockout time of not more than 90 seconds as specified in 3.5.3.2 shall constitute failure of this test.

4.5.9 Flame signal test. An ac-dc multimeter with a 20,000 ohm/volt dc rating shall be connected to the test jacks of the flame safeguard control (UVM-2). When flame is detected a steady dc 5 to 6 volt should be obtained. When the flame is extinguished and the spark is on, the reading should be 0 volts and within 2 seconds. Any nonconformance with the control requirements of 3.5.3.3 shall constitute failure of this test.

4.5.10 Signal device test. The buzzer shall sound using the test in 4.5.4 and 4.5.8. Any nonconformance with the requirements of 3.5.4 shall constitute failure of this test.

4.5.11 Circuit breaker test. All circuit breakers shall operate as intended without damage to themselves or other electrical component. Any nonconformance with the requirements of 3.5.5 shall constitute failure of this test.

* 4.5.12 Main circuit breaker test. A current of 8 amps shall be applied to the load side of the circuit breaker. The circuit breaker shall open within a 3 minute period and after 5 minutes of cooling the breaker should be reset by closing the switch. Any nonconformance with the requirements of 3.5.3.1 shall constitute failure of this test.

* 4.5.13 Line starter test. A current of 8 amps shall be applied to the load side of the circuit breaker. The circuit breaker shall open within a 3 minute period and after 5 minutes of cooling the breaker shall be reset by closing the switch. Any nonconformance with the requirements of 3.5.3.2 shall constitute failure of this test.

4.5.14 Burner, motor test. A locked rotor test shall be performed to determine if the electrical power to the burner motor shut off with 25 seconds. After 5 minutes of cooling the overload switch shall be reset by depressing the button. Any nonconformance to the requirements as specified in 3.5.5.3 shall constitute failure of this test.

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4.5.15 Relief valve test. The relief valve shall be tested for opening and closing. Any nonconformance with the requirements of 3.5.6 shall constitute failure of this test.

4.5.16 Burner, smoke test. The heater burner shall be operated and checked for visible smoke. Any nonconformance with the requirements of 3.5.7 shall constitute failure of this test.

4.5.17 Heater capacity test. With the operating control by-passed, and a constant flow of 9 gallons per minute thru the heater, the inlet and outlet temperatures shall be recorded. Temperature rise exceeding 100°F shall be maintained once the maximum temperature has been reached. Any nonconformance with the requirements of 3.5.8 shall constitute failure of this test.

4.5.18 Pump fuel test. The fuel pump gage shall read a pressure of 100 psi ± 2 gage anytime during burner operation. Any nonconformance with the requirements of 3.5.9 shall constitute failure of this test.

4.5.19 Heater assembly test. One heater assembly shall be placed in operation. Any nonconformance with the assembly and operation requirements of 3.5.10 shall constitute failure of this test.

4.5.20 Leakage test. A dry air pressure of 50 psi gage shall be applied to all hose assemblies and the components checked for leaks. Any nonconformance with the leakage requirements of 3.5.11 shall constitute failure of this test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Drainage of water system. The entire water heater shall be thoroughly drained, blown out and dried. All valves and draincocks shall be left open.

* 5.1.1.2 Fuel system. The fuel supply and return lines shall be connected to the oil filter and fuel pump, respectively. The hose shall be coiled beneath the heater and the ends connected to the fuel line holder. The burner pump shall be filled with P-10, grade 10 or grade 30 of MIL-P-116.

* 5.1.1.3 Cleaning, drying, and preservative application. All uncoated and unpainted surfaces of the heater shall be cleaned in accordance with MIL-P-116, process C-1, and thoroughly dried prior to application of preservative. Preservative specified in 5.1.1.5 shall conform to the applicable specification listed in MIl-P-116. Barrier material specified in 5.1.1.5 shall conform to MIL-B-121, type I or II, grade A, class 2. Tape specified hereinafter shall conform to type II of MIL-T-22085.

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5.1.1.4 Repainting and recoating. Painted and permanently coated surfaces from which the paint or permanent coating has been removed or which are defective, shall be repainted or recoated with material identical with the original.

5.1.1.5 Heater assembly. The heater assembly shall be drained and dried of water and fuel as specified in 5.1.1.1. All interior surfaces that are unlined, or not treated to resist corrosion, shall be coated with P-10, type I, grade 30 preservative applied by fogging, spraying, or completely filling and draining. Excess preservative shall be drained and the drain and fill plugs, before being reinstalled, shall be coated with the preservative. All water line openings on the heater assembly shall be sealed with tape. All openings on the air shutter, smoke stack and electric motor shall be sealed with barrier material and tape, or by tape only, as applicable. All gages, dials, and sight glasses not protected by a cover, shall be cushioned, overwrapped with barrier material, and the wrap secured in place with tape.

5.1.1.6 Drum fill adapter assembly. The drum adapter extension shall be threaded, hand-tight into the 1/4 inch return port.

* 5.1.2 Commercial. The heater shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A or Commercial as specified (see 6.2).

5.2.1 Level A packing.

* 5.2.1.1 Heater assembly and lower smoke stack assembly. The heater, and drum fill adapter, preserved as specified in 5.1, shall be packed in snug-fitting nailed wood box constructed in accordance with PPP-B-621, class 2, grade A, style 7, type 3 load. The water heater shall be bolted to the box bottom through the skids. Additional skids are required and shall be located on the box bottom adjacent to the inner edges of the outside skids to allow for bolting of the water heater through the skids. The elbow and lower stove pipe section of the smoke stack assembly shall be attached to the boiler. The fuel lines shall be coiled beneath the heater and the ends attached to the fuel line holder. Assembly of the box shall be in accordance with the appendix of PPP-B-621, except that the sheeting on the ends shall be attached to the interior frame members with nails and to the side of the skids with screws. The exterior surfaces of the assembled box shall be painted with one coat of primer conforming to TT-W-572 and one coat of enamel conforming to class A of TT-E-529. The color of the enamel shall be olive drab number 34087 of FED-STD-595. The assembled and painted box shall be strapped as specified in the appendix of PPP-B-621.

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* 5.2.2 Commercial packing. The heater preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.3.1 Additional marking. For level A shipping containers as specified in 5.2.1.1, the following additional marking shall be printed or stenciled in orange stencil ink conforming to type TT-I-1795, type II, on the top of each container in block characters not less than 1 inch high:

REUSABLE CONTAINER

DO NOT DESTROY

TO OPEN REMOVE SCREWS

6. NOTES

6.1 Intended use. The water heater is to provide hot water for the military field sanitation, laundry and bath units.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. When a first article is required (see 3.2, 4.3 and 6.3).
c. When certification of qualification test for welders is not required (see 3.8.1).
d. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample consisting of one heater. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.3).

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* 6.5 Changes from previous issue. The margins of this document have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD

Preparing activity:

Army - GL

Project No. 4520-0290

Review activities:

Army - MD
DLA - CS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-44086A	Heater, Water, Liquid Fuel; M-80

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP, 1 Apr 84

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